

AC Voltage Sensor:

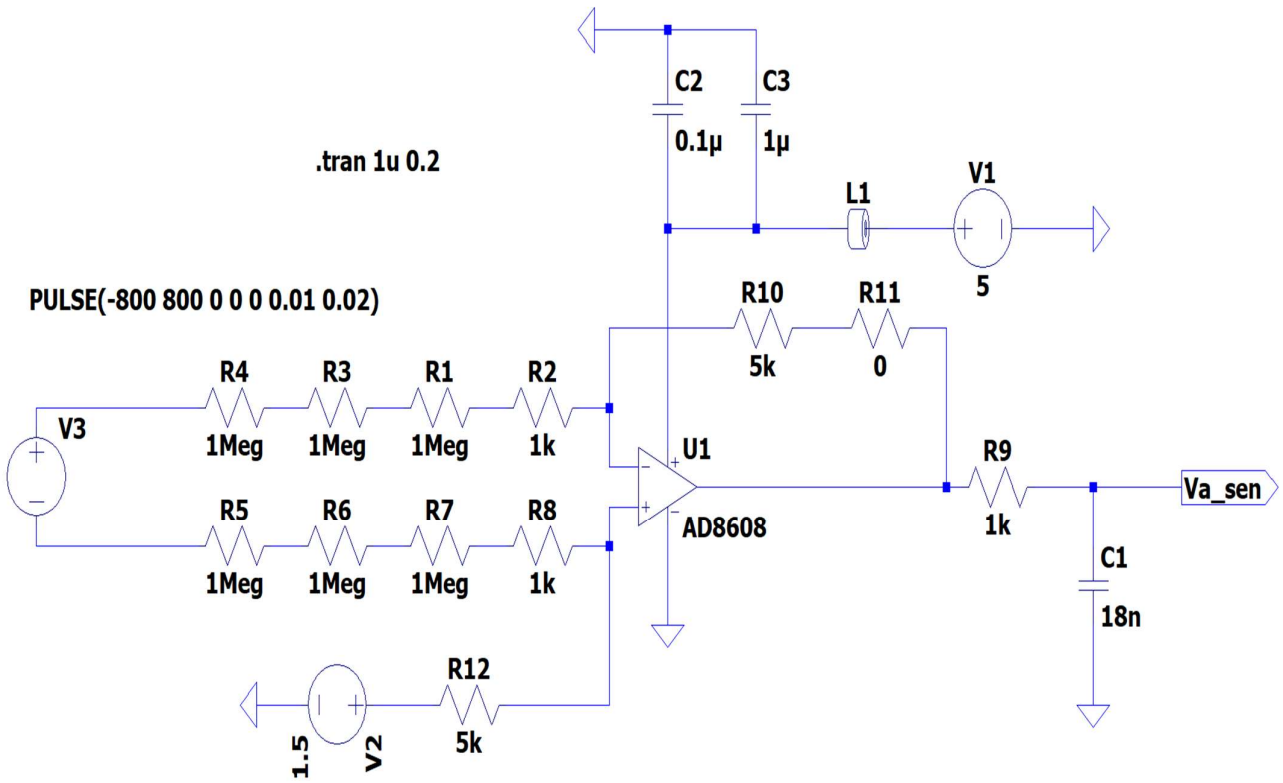


Fig- LTSpice schematics of one phase of the AC Voltage Sensor

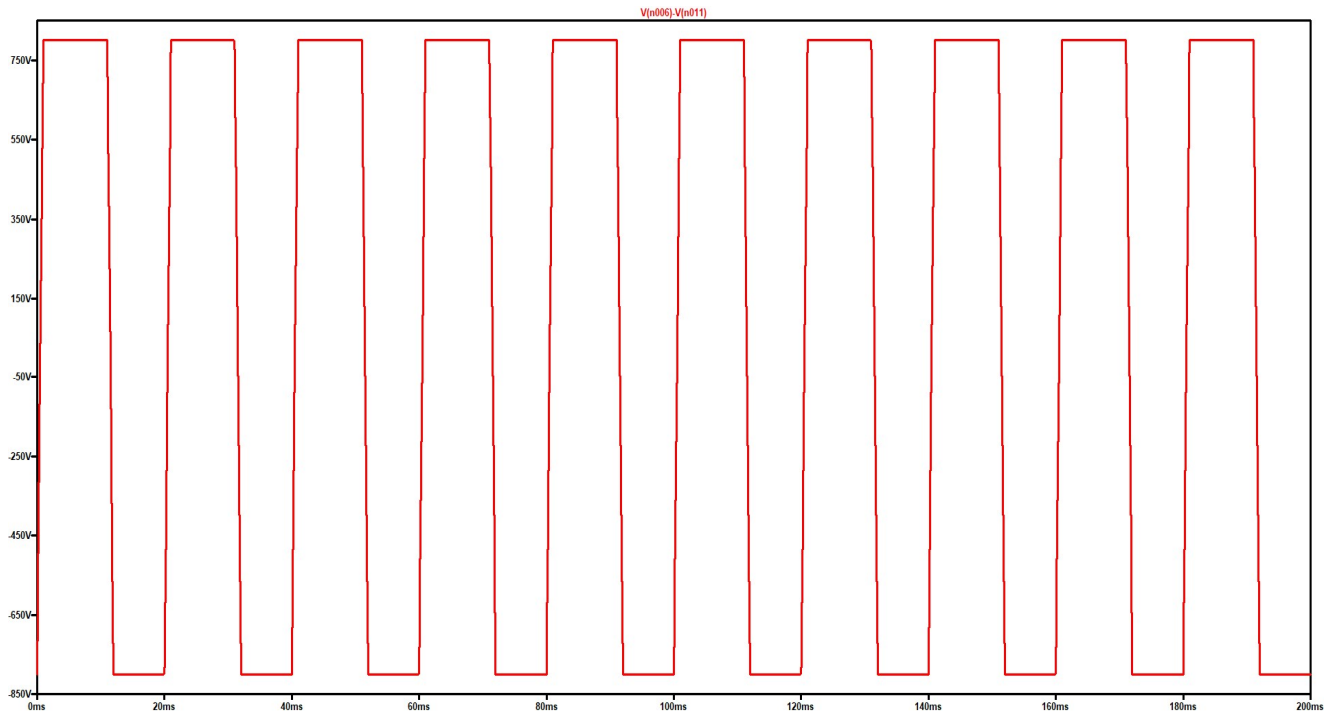


Fig- Input voltage to the sensor

- Here it is a square wave of 1600V (peak-to-peak) at a frequency of 50Hz.

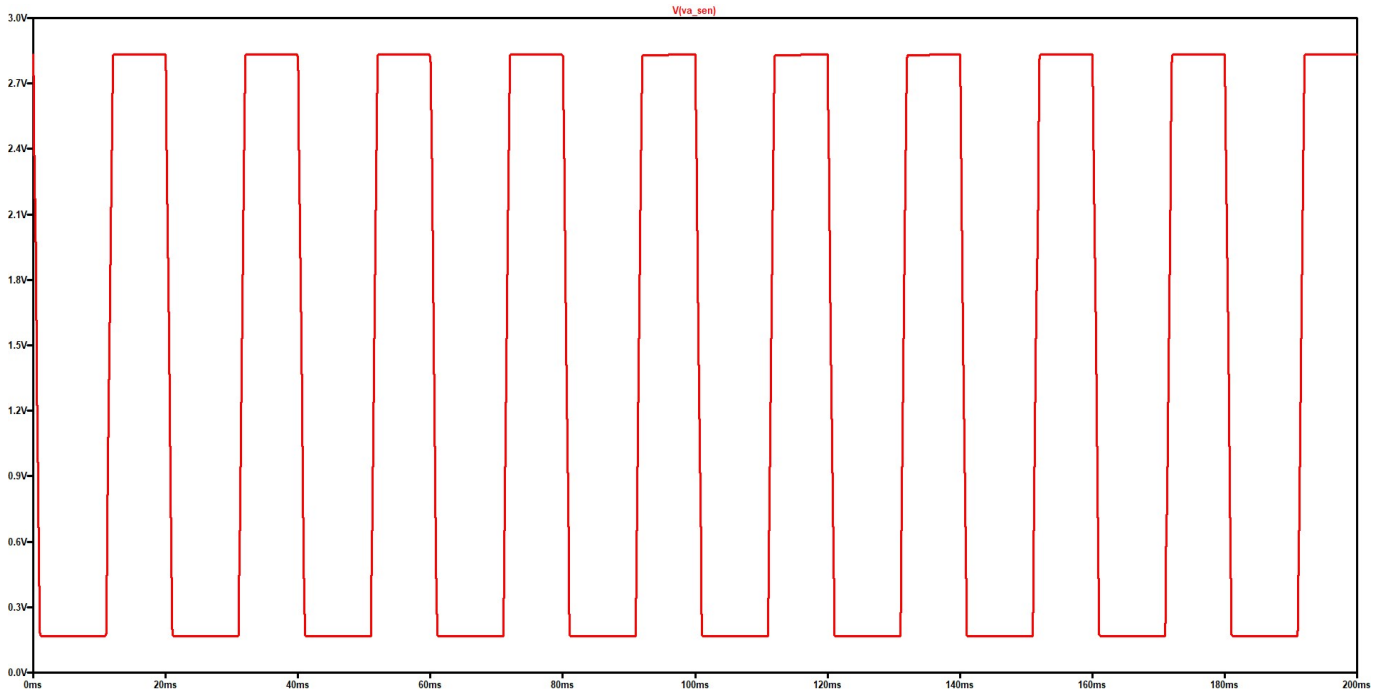


Fig- Output of AC Voltage sensor

From LTSpice we get,

Here it is a square wave of peak-to-peak voltage of $2.833 - 0.166 = 2.667$

$$\text{Gain} = \frac{1600}{2.667} = 599.925$$

DC Voltage Sensor:

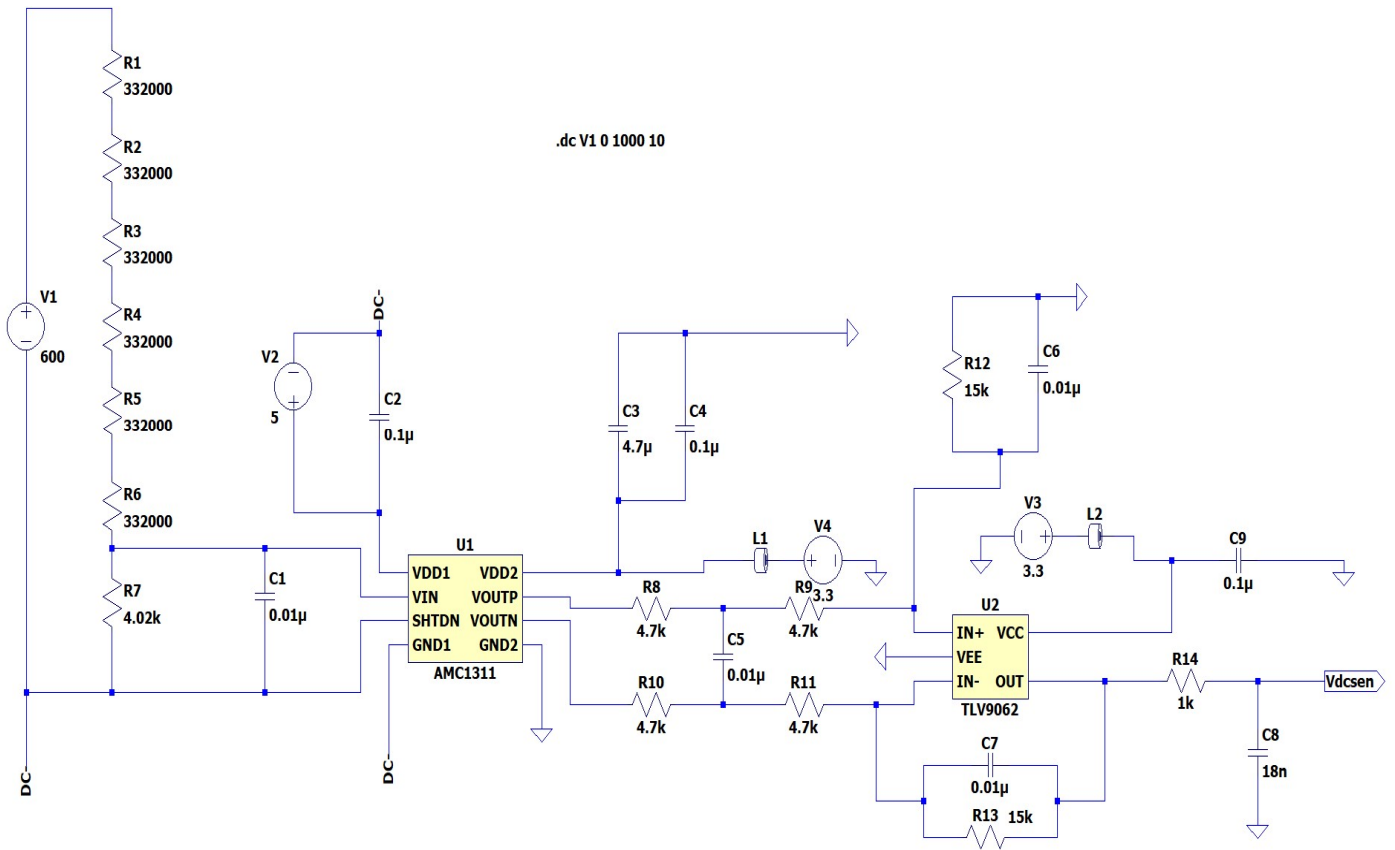


Fig- LTSpice schematics of the DC Voltage Sensor

- The DC input voltage is swept from 0V to 1000V with steps of 10V.
- The resistor divider circuit given brings down the voltage, so that it can act as an appropriate input for the AMC1311 IC.

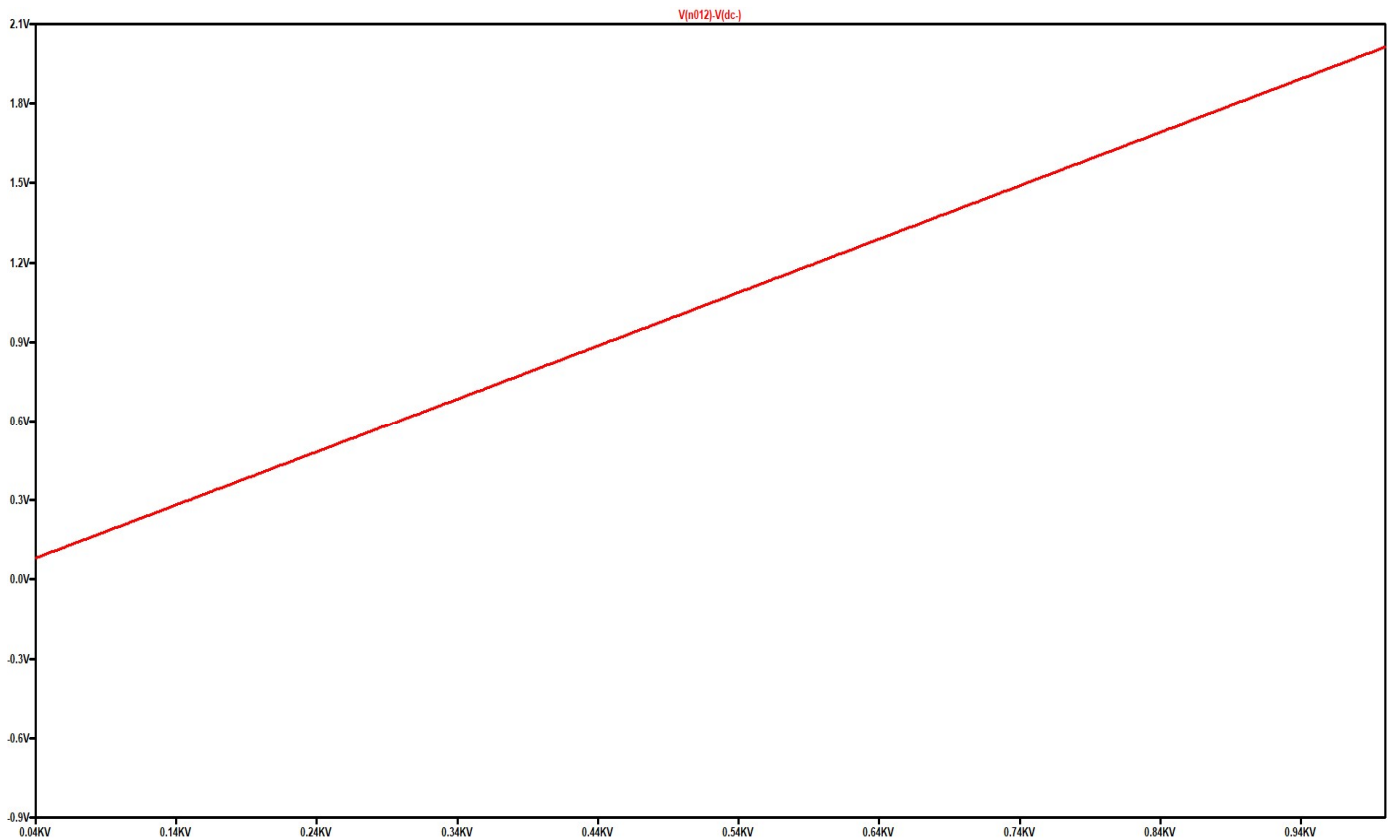


Fig- Voltage waveform across the capacitor C1

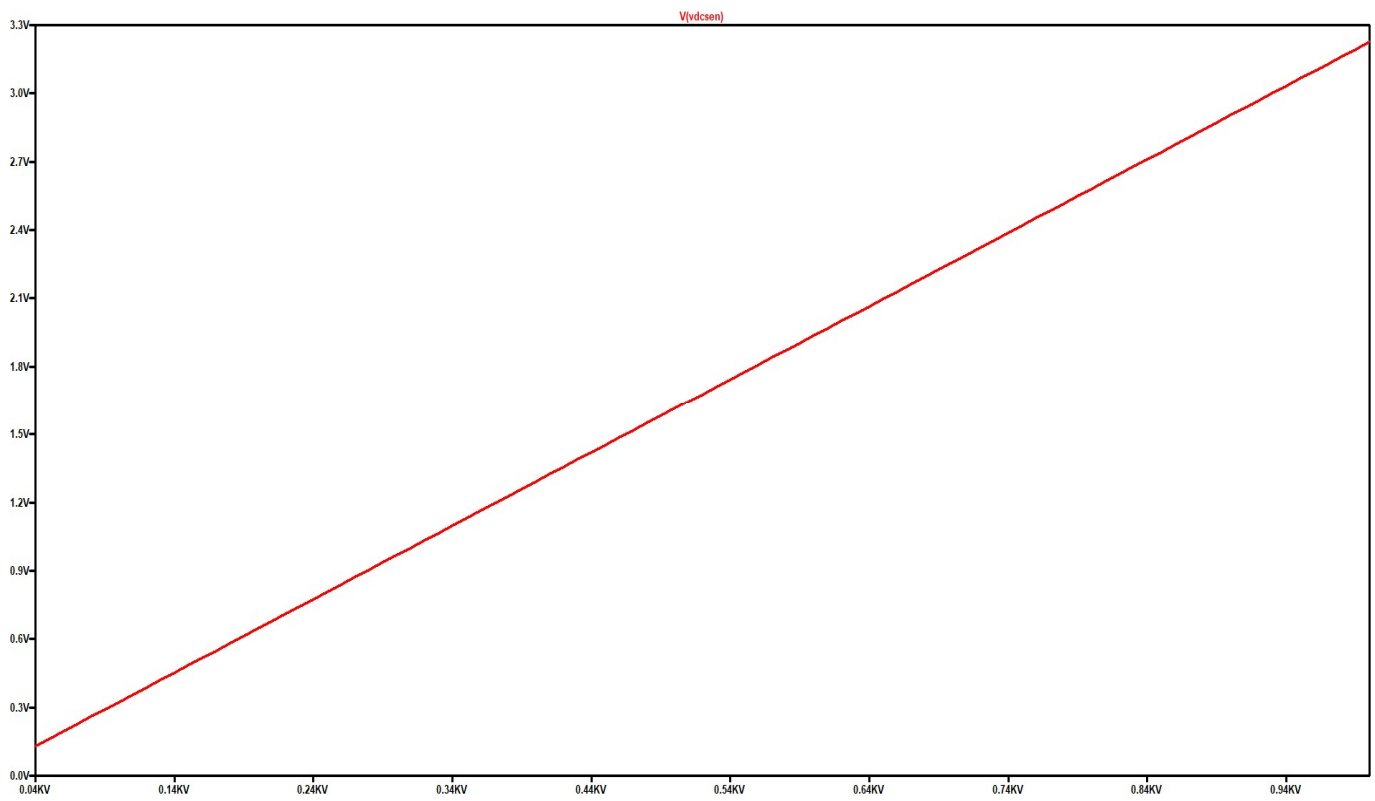


Fig- Output of DC Voltage sensor

Current Sensor:

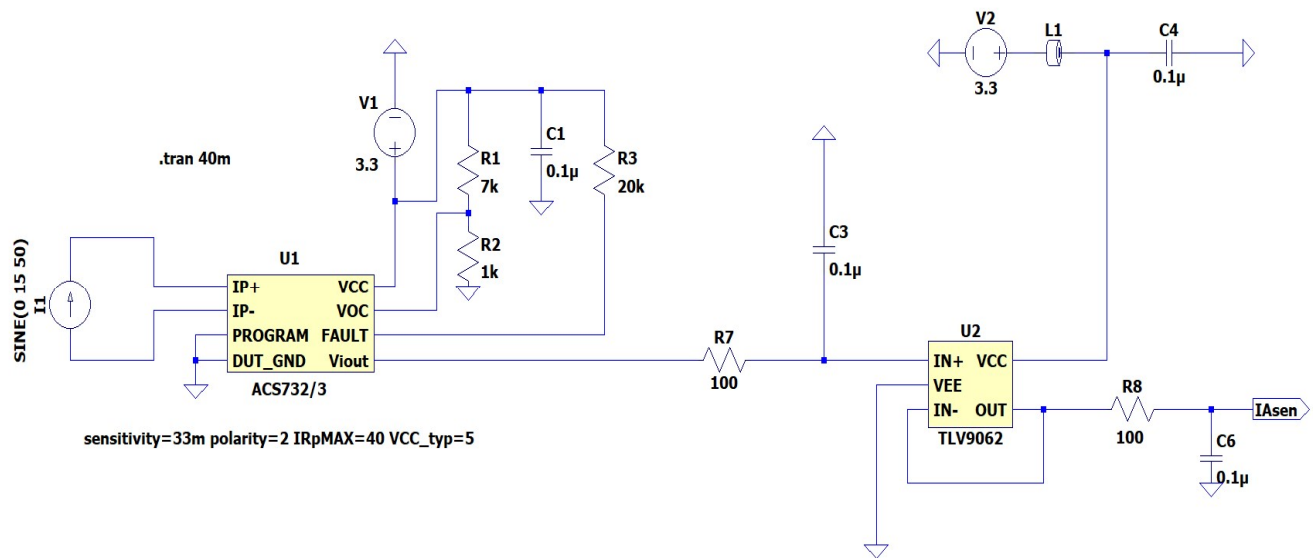


Fig- LTSpice schematics of the Current Sensor

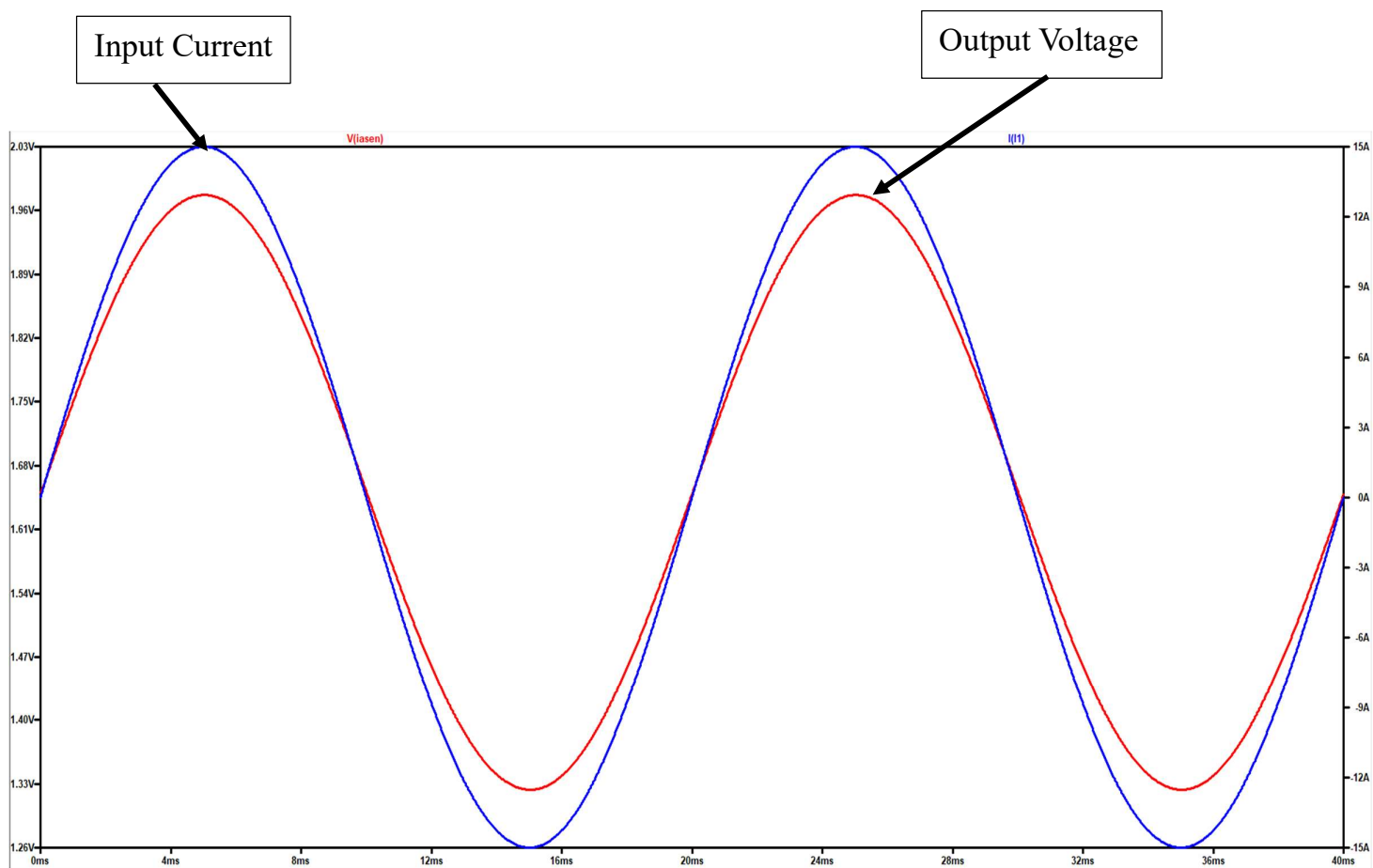


Fig- Input Current and Output sensed voltage waveforms

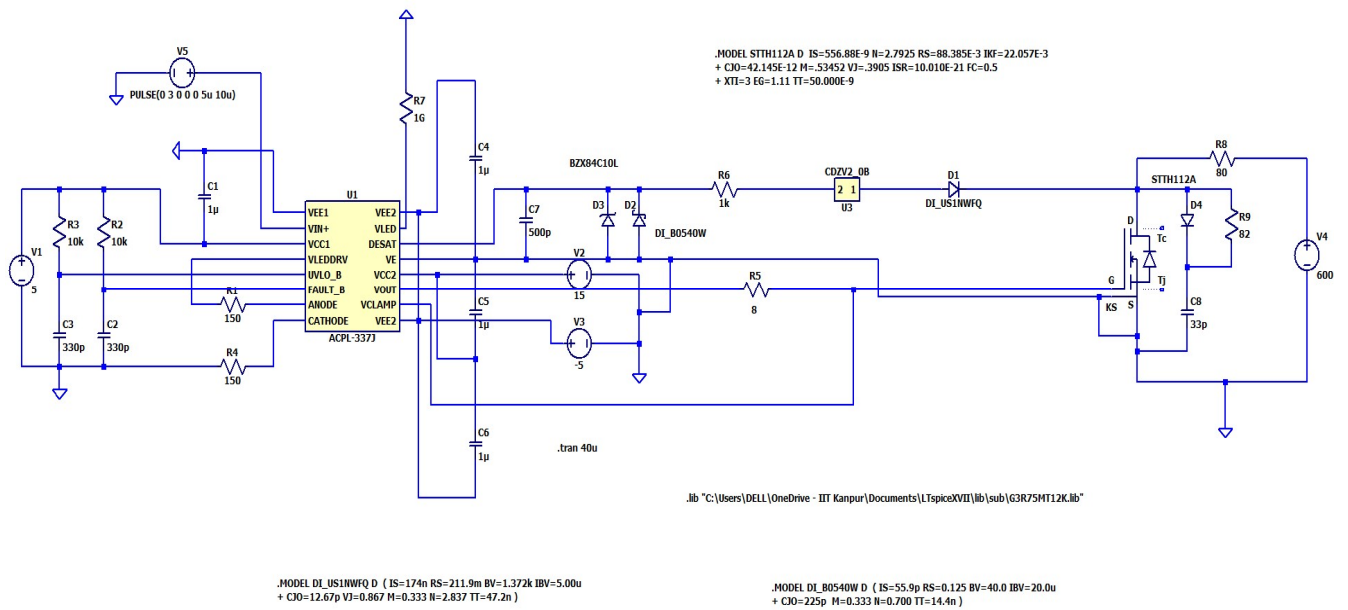


Fig- LTSpice schematic of Gate Driver- ACPL337J

